

DBMS Lab Assignment – 1

Name :- Parth Bhayani PRN NO :- 26070122512 Batch :- B3

Title: Concept of E-R and EER Diagrams and Conversion to Relational Schema

Description:

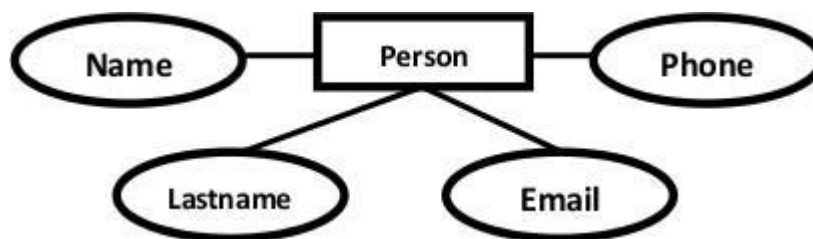
The ER Model is intended as a description of real-world entities. Although it is constructed in such a way as to allow easy translation to the relational schema model, this is not an entirely trivial process. The ER diagram represents the conceptual level of database design meanwhile the relational schema is the logical level for the database design. We will be following the simple rules:

1. Entities and Simple Attributes:

An entity type within ER diagram is turned into a table. You may preferably keep the same name for the entity or give it a sensible name but avoid DBMS reserved words as well as avoid the use of special characters. Each attribute turns into a column (attribute) in the table. The key attribute of the entity is the primary key of the table which is usually underlined. It can be composite if required but can never be null.

It is highly recommended that every table should start with its primary key attribute conventionally named as `TablenameID`.

Taking the following simple ER diagram:



The initial relational schema is expressed in the following format writing the table names with the attributes list inside a parentheses as shown below for

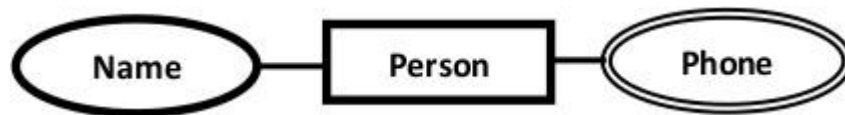
Persons(personid , name, lastname, email)

Persons and Phones are Tables. name, lastname, are Table Columns (Attributes).

personid is the primary key for the table : Person

2. Multi-Valued Attributes

A multi-valued attribute is usually represented with a double-line oval.



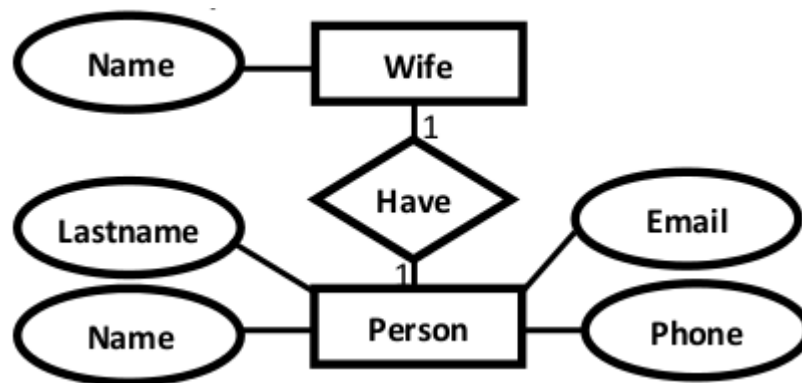
If you have a multi-valued attribute, take the attribute and turn it into a new entity or table of its own. Then make a 1:N relationship between the new entity and the existing one. In simple words. 1. Create a table for the attribute. 2. Add the primary (id) column of the parent entity as a foreign key within the new table as shown below:

Persons(personid , name, lastname, email)

Phones (phoneid , personid, phone)

personid within the table Phones is a foreign key referring to the personid of Persons

3. 1:1 Relationships



To keep it simple and even for better performances at data retrieval, I would personally recommend using attributes to represent such relationship. For instance, let us consider the case where the Person has or optionally has one wife. You can place the primary key of the wife within the table of the Persons which we call in this case Foreign key as shown below.

**Persons(personid , name, lastname, email ,
wifeid)** **Wife (wifeid , name)**

Or vice versa to put the **personid** as a foreign key within the Wife table as shown below:

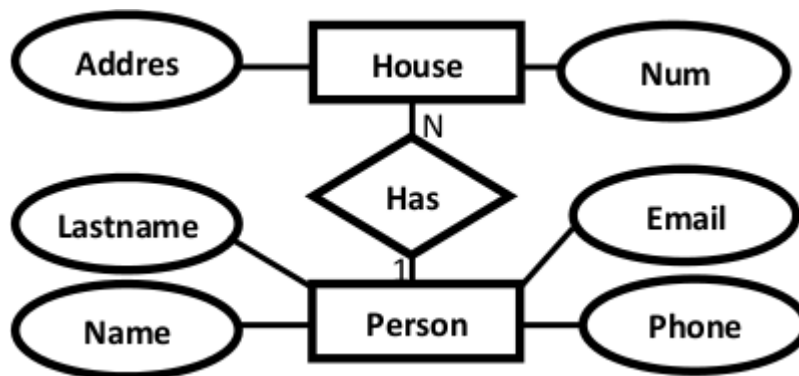
**Persons(personid , name, lastname,
email)** **Wife (wifeid , name , personid)**

For cases when the Person is not married i.e. has no wifeID, the attribute can set to NULL

4. 1:N Relationships

This is the tricky part ! For simplicity, use attributes in the same way as 1:1 relationship but we have only one choice as opposed to two choices. For instance, the Person can have a House from

zero to many , but a House can have only one Person. To represent such relationship the personid as the Parent node must be placed within the Child table as a foreign key but not the other way around as shown next:



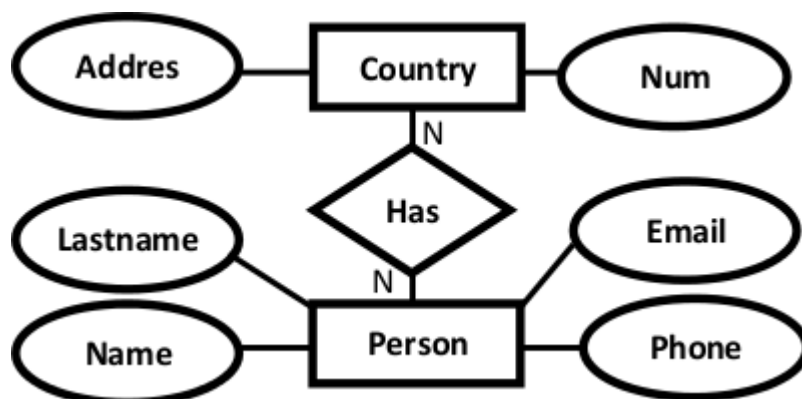
It should convert to :

Persons(personid , name, lastname, email)

House (houseid , num , address, *personid*)

5. N:N Relationships

We normally use tables to express such type of relationship. This is the same for N – ary relationship of ER diagrams. For instance, The Person can live or work in many countries. Also, a country can have many people. To express this relationship within a relational schema we use a separate table as shown below:



It should convert into :

Persons(personid , name, lastname, email)

Countries (countryid , name, code)

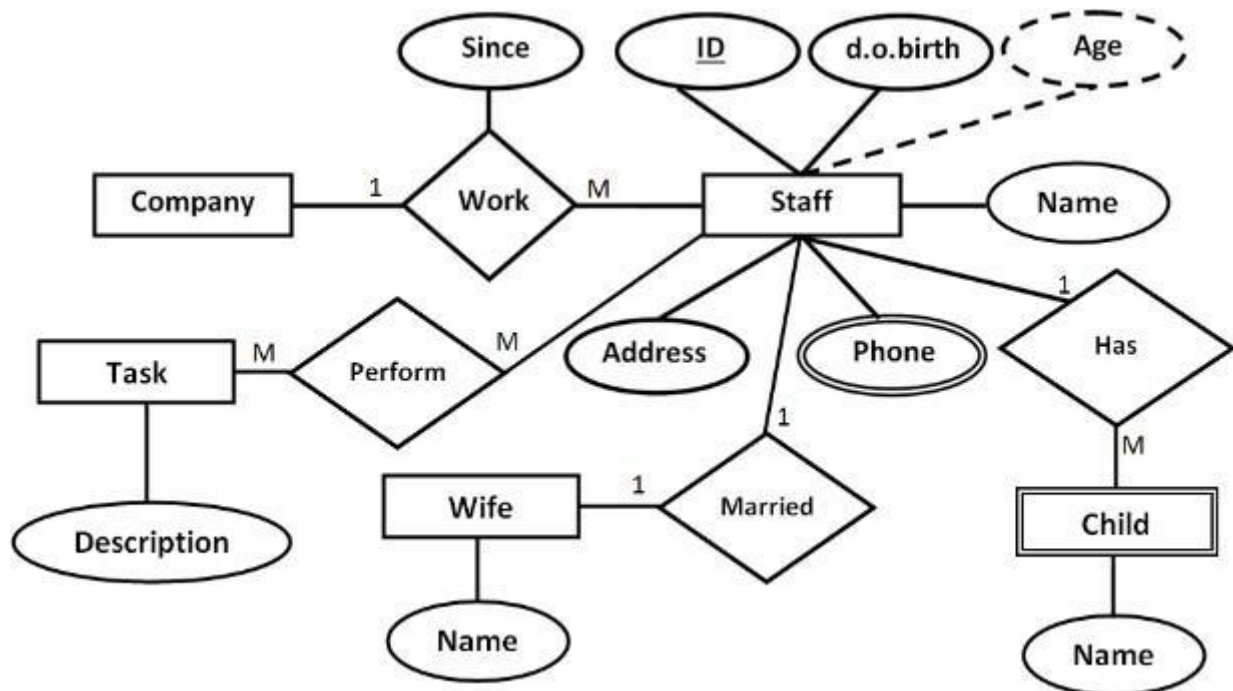
HasRelat (hasrelatid , personid , countryid)

Relationship with attributes:

It is recommended to use table to represent them to keep the design tidy and clean **regardless of the cardinality** of the relationship.

Practice Example

Convert the following ER diagram into database relational schema:



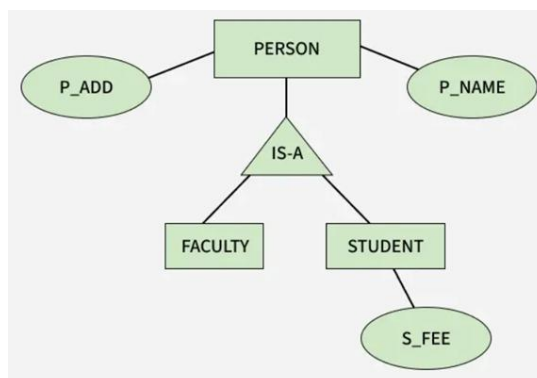
The relational schema for the ER Diagram is given below as:

Company(CompanyID , name , address)
Staff(StaffID , dob , address , WifeID)
Child(ChildID , name , StaffID)
Wife (WifeID , name)
Phone(PhoneID , phoneNumber , StaffID)
Task (TaskID , description)
Work(WorkID , CompanyID , StaffID , since)
Perform(PerformID , StaffID , TaskID)

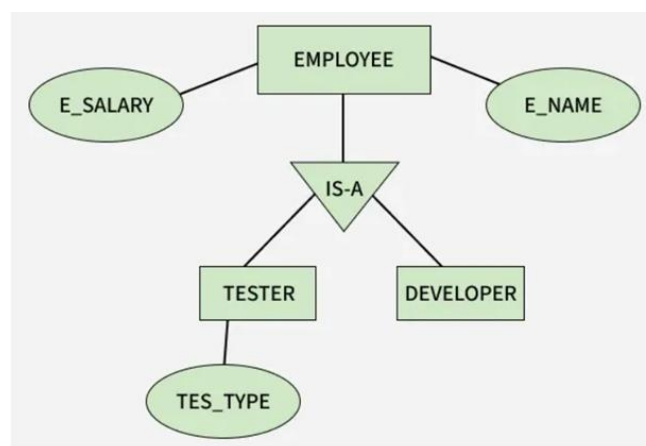
Conversion of EER Model into Relational Model

The Extended Entity-Relationship Model is a more complex and high-level model that extends an E-R diagram to include more types of abstraction, and to more clearly express constraints. All of the concepts contained within an E-R diagram are included in the EE-R model, along with additional concepts that cover more semantic information. These additional concepts include generalization, specialization, and aggregation.

Generalization is a bottom-up approach in which two lower level entities combine to form a higher level entity. In generalization, the higher level entity can also combine with other lower level entity to make further higher level entity.



Specialization is opposite to Generalization. It is a top-down approach in which one higher level entity can be broken down into two lower level entities. In specialization, some higher level entities may not have lower-level entity sets at all.



Conversion process for Generalization/ Specialization into Relational Schema:

- Create tables for all higher-level entities.
- Create tables for lower-level entities.
- Add primary keys of higher-level entities in the table of lower-level entities.
- In lower-level tables, add all other attributes of lower-level entities.
- Declare primary key of higher-level table and the primary key for lower-level table.
- Declare foreign key constraints.

Representing Generalization (Approach #1)

Create a table for each lower-level entity set only

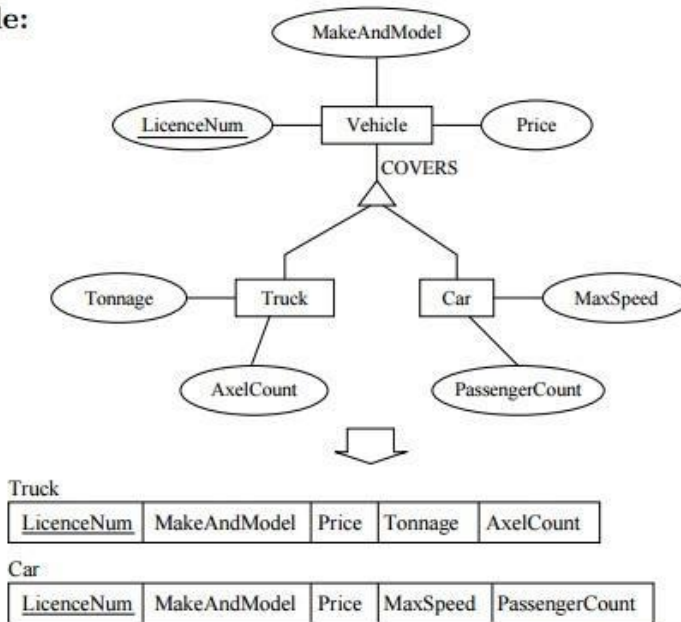
Columns of new tables should include

- Attributes of lower level entity set
- Attributes of the superset

The higher-level entity set can be defined as a view on the tables for the lower-level entity sets

Representing Generalization (Approach #1)

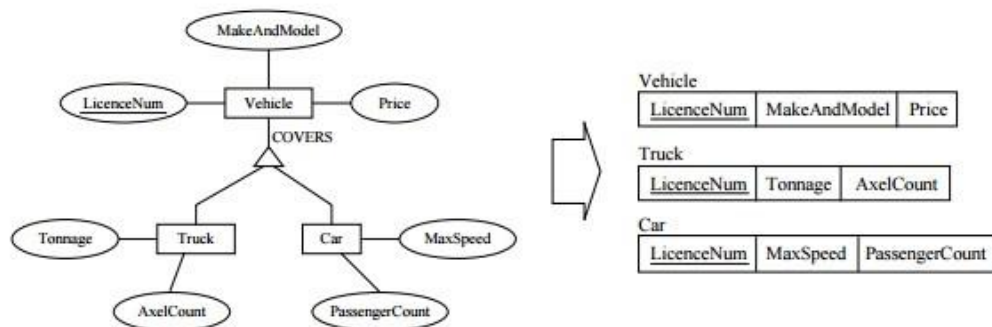
Example:



Representing Generalization (Approach #2)

Treat generalization the same as specialization.

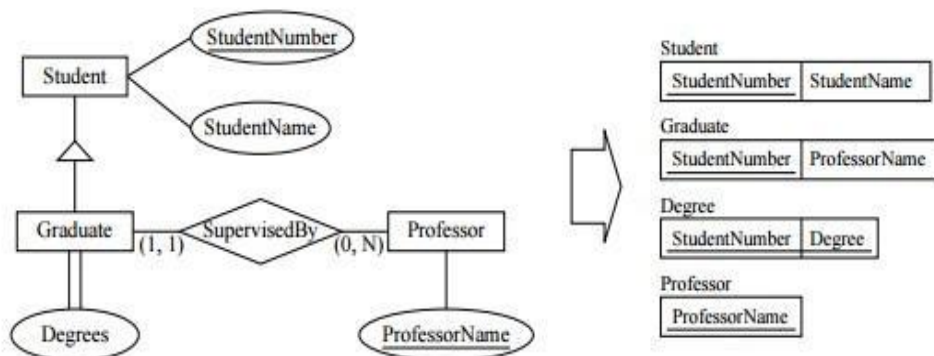
Example:



Representing Specialization

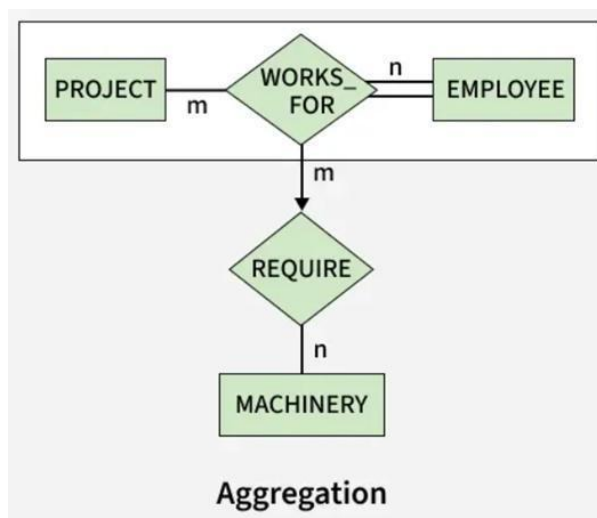
Create table for higher-level entity set, and treat specialized entity subsets like weak entity sets (without discriminators)

Example:



One limitation on ER model is that it cannot express relationships among relationships.

Aggregation provides best solution for this purpose. Aggregation is a process when relation between two entities is treated as a single entity. To represent relationship set involving aggregation of R, treat the aggregation like an entity set whose primary key is the primary key of the table for R.



Assignment 1 Lab Questions

Theory Questions

1. What are ER Diagrams? What is their significance in DBMS?

An **Entity-Relationship (ER) Diagram** is a graphical representation of the structure of a database. It illustrates the **entities (objects or things)**, **attributes (properties)**, and **relationships** between entities. ER diagrams are used during the database design process to create a clear blueprint before implementing the database.

Main Components of an ER Diagram:

- **Entity:** A real-world object or concept (e.g., Student, Employee, Book).
- **Attribute:** A characteristic of an entity (e.g., Student_ID, Name, Age).
- **Relationship:** The association between two or more entities (e.g., A Student enrolls in a Course).

Significance of ER Diagrams in DBMS

ER diagrams are important in a **Database Management System (DBMS)** because they:

1. **Provide a clear visual representation** of the database structure.
2. **Help in database design** by identifying entities, attributes, and relationships before implementation.
3. **Reduce data redundancy** by organizing data efficiently.
4. **Improve communication** between database designers, developers, and users through an easy-to-understand model.
5. **Ensure data integrity** by defining relationships and constraints clearly.
6. **Simplify database maintenance** and future modifications.
7. **Serve as a blueprint** for converting the conceptual design into relational database tables.

2. Explain the conversion process from ER to Relational.

The **ER-to-Relational conversion** is the process of transforming an **Entity-Relationship (ER) Diagram** into a **Relational Database Schema (tables)**. During this process, entities, attributes, and relationships are converted into tables with appropriate keys and constraints.

Steps in ER to Relational Conversion

1. **Convert Strong Entities into Tables**
 - Create a separate table for each strong entity.

- Entity attributes become table columns.
- The primary key of the entity becomes the table's primary key.

2. Convert Weak Entities into Tables

- Create a separate table for each weak entity.
- Include the primary key of the owner entity as a foreign key.
- The combined key forms the primary key of the weak entity table.

3. Convert One-to-One (1:1) Relationships

- Add the primary key of one entity as a foreign key in the other entity's table.
- Place the foreign key in the table with total participation whenever possible.

4. Convert One-to-Many (1:M) Relationships

- Add the primary key of the "one" side entity as a foreign key in the "many" side entity's table.

5. Convert Many-to-Many (M:N) Relationships

- Create a separate table for the relationship.
- Include the primary keys of both participating entities as foreign keys.
- The combination of these keys forms the primary key of the relationship table.
- Include any relationship attributes in this table.

6. Convert Multivalued Attributes

- Create a separate table for each multivalued attribute.
- Include the primary key of the parent entity as a foreign key.

7. Convert Composite Attributes

- Replace each composite attribute with its individual simple attributes.
- Store only the simple attributes in the table.

3. Explain the need of EER diagrams

Enhanced Entity-Relationship (EER) Diagrams are an extension of ER diagrams that provide additional features to model complex database systems. They include concepts such as **specialization, generalization, inheritance, aggregation, and categories**, making database design more expressive and flexible.

Need of EER Diagrams

1. Model Complex Databases

- EER diagrams can represent complex real-world scenarios that cannot be modeled effectively using basic ER diagrams.

2. Support Inheritance

- They allow entities to inherit attributes and relationships from parent entities through specialization and generalization.

3. Reduce Data Redundancy

- Common attributes can be stored in a superclass, avoiding duplication in multiple related entities.

4. Improve Database Design

- They provide a more detailed and organized representation of the database structure.

5. Represent Real-World Relationships

- EER diagrams accurately model hierarchical and complex relationships among entities.

6. Enhance Flexibility

- They make it easier to modify and extend the database as requirements change.

7. Improve Clarity and Maintenance

- A well-structured EER diagram is easier to understand, implement, and maintain.

4. Explain in detail Generalization, specialization and Aggregation and how do we convert them into relational model

1. Generalization

Generalization is the process of combining two or more similar lower-level entities into a single higher-level entity called a **superclass**. The superclass contains the common attributes shared by all the lower-level entities.

Features:

- It is a **bottom-up approach**.
- Reduces data redundancy by storing common attributes in one place.
- The superclass represents the generalized form of all subclasses.

Conversion into Relational Model

- Create one table for the **superclass** containing the common attributes.
- Create separate tables for each **subclass** containing their specific attributes.
- The primary key of the superclass becomes the primary key as well as the foreign key in each subclass table.

2. Specialization

Specialization is the process of dividing a higher-level entity into two or more lower-level entities called **subclasses** based on specific characteristics. Each subclass inherits the attributes of the superclass and may have additional attributes.

Features:

- It is a **top-down approach**.
- Supports inheritance of attributes and relationships.
- Represents more specific entities from a general entity.

Conversion into Relational Model

- Create one table for the **superclass** with all common attributes.
 - Create separate tables for each **subclass** with their unique attributes.
 - The primary key of the superclass is used as the primary key and foreign key in each subclass table.
-

3. Aggregation

Aggregation is a process in which a relationship between entities is treated as a higher-level entity. It is used when a relationship itself participates in another relationship.

Features:

- Represents a **relationship as an entity**.
- Helps model complex real-world situations.
- Simplifies the representation of higher-level relationships.

Conversion into Relational Model

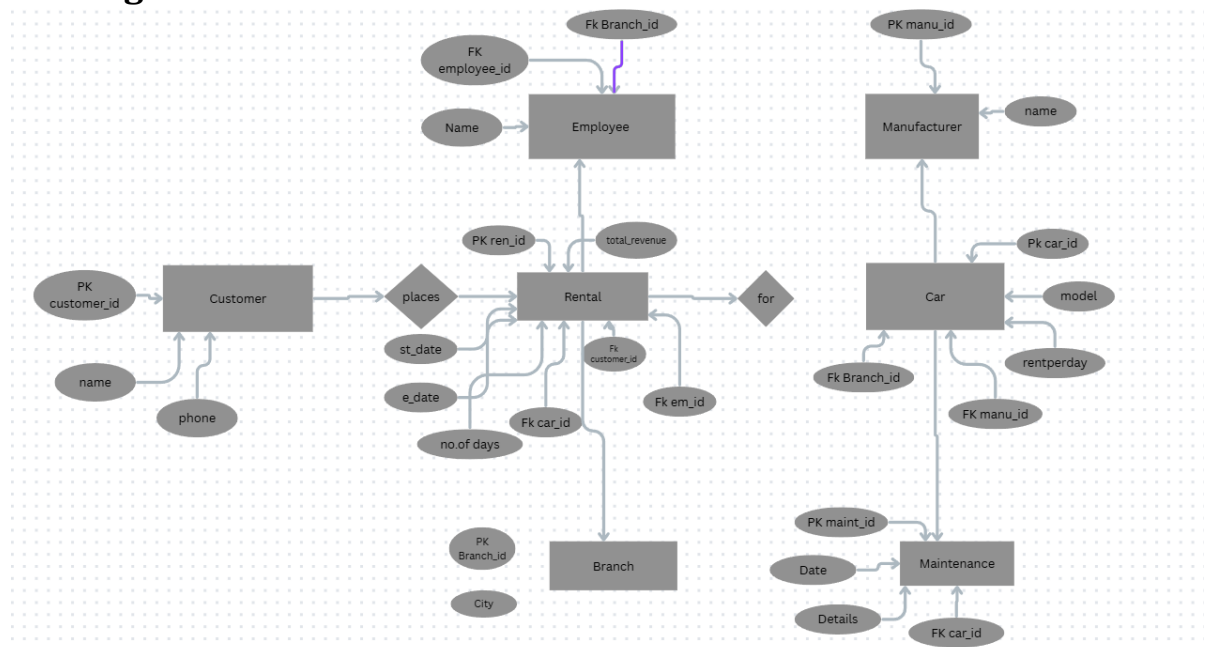
- Create tables for all participating entities.
- Create a separate table for the aggregated relationship.
- Include the primary keys of the participating entities as foreign keys in the aggregation table.
- Add any attributes belonging to the aggregated relationship.

Implementation Question (ER):- Draw an ER diagram for the following problem statements and then convert them into database relational schema

1) CAR Rental Management

- One customer can rent one car.
- Each car has one manufacturer.
- Each manufacturer can manufacture many types of cars.
- Each car has its own maintenance schedule.
- The revenue is generated based on number of days the car was rented.
- The company has branches in 5 major cities.
- The companies also have 5 employees per branch.
- Initial rental request are handled by employees.

ER Diagram



Relational Schema :-

CUSTOMER(Customer_ID PK, Name, Phone)

BRANCH(Branch_ID PK, City)

EMPLOYEE(Employee_ID PK, Name, Branch_ID FK)

MANUFACTURER(Manu_ID PK, Name)

CAR(Car_ID PK, Model, RentPerDay, Branch_ID FK, Manu_ID FK)

RENTAL(Ren_ID PK, Start_Date, End_Date, No_of_Days, Total_Revenue, Customer_ID FK, Employee_ID FK, Car_ID FK)

MAINTENANCE(Maint_ID PK, Date, Details, Car_ID FK)

Relational Model :-

CUSTOMER		
Customer_ID (PK)	Name	Phone
C001	Rahul Sharma	9876543210
C002	Priya Patel	9123456780
C003	Amit Verma	9988776655

BRANCH	
Branch_ID (PK)	City
B001	Pune
B002	Mumbai
B003	Nashik

EMPLOYEE		
Employee_ID (PK)	Name	Branch_ID (FK)
E001	Rohan Patil	B001
E002	Sneha Shah	B002
E003	Karan Mehta	B003

MANUFACTURER	
Manu_ID (PK)	Name
M001	Maruti Suzuki
M002	Hyundai
M003	Tata Motors

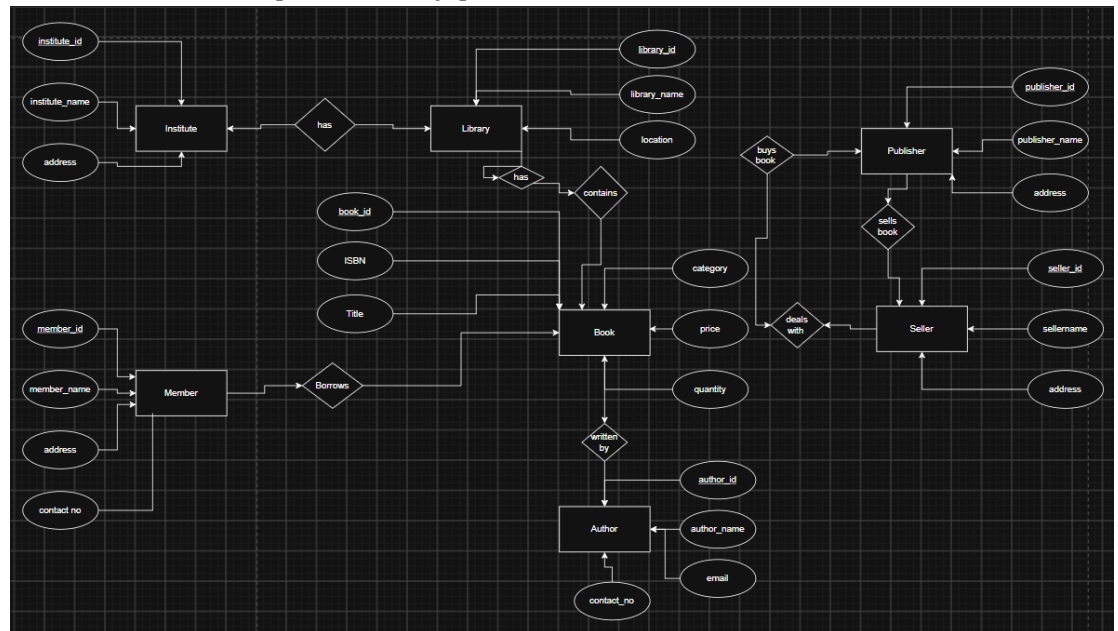
CAR				
Car_ID (PK)	Model	RentPerDay	Branch_ID (FK)	Manu_ID (FK)
CR001	Swift	1800	B001	M001
CR002	Creta	2800	B002	M002
CR003	Nexon	2500	B003	M003

RENTAL							
Ren_ID (PK)	Start_Date	End_Date	No_of_Days	Total_Revenue	Customer_ID (FK)	Employee_ID (FK)	Car_ID (FK)
R001	01-07-2026	05-07-2026	5	9000	C001	E001	CR001
R002	08-07-2026	10-07-2026	3	8400	C002	E002	CR002
R003	12-07-2026	15-07-2026	4	10000	C003	E003	CR003

MAINTENANCE			
Maint_ID (PK)	Date	Details	Car_ID (FK)
MT001	20-07-2026	Engine Service	CR001
MT002	22-07-2026	Oil Change	CR002
MT003	25-07-2026	Tyre Replacement	CR003

2) SIU Library Management System.

- SIU library there can be individual institute libraries.
- Library has members.
- Library buys books of many publishers.
- Sellers sell books of many publishers.
- Library contains books
- Books are written by one/many authors.
- Authors' books are published by publishers.



Relational Schema :-

- INSTITUTE(Institute_ID PK, Institute_Name, Address)
- LIBRARY(Library_ID PK, Library_Name, Location, Institute_ID FK)
- MEMBER(Member_ID PK, Member_Name, Address, Contact_No)
- AUTHOR(Author_ID PK, Author_Name, Email, Contact_No)
- PUBLISHER(Publisher_ID PK, Publisher_Name, Address)
- SELLER(Seller_ID PK, Seller_Name, Address, Publisher_ID FK)
- BOOK(Book_ID PK, ISBN, Title, Category, Price, Quantity, Library_ID FK, Author_ID FK, Publisher_ID FK, Seller_ID FK)
- BORROWS(Member_ID PK/FK, Book_ID PK/FK)

Relational Model :-

INSTITUTE		
Institute_ID (PK)	Institute_Name	Address
I001	Symbiosis International University	Lavale, Pune
I002	Engineering Campus	Lavale, Pune
I003	Management Campus	Lavale, Pune

LIBRARY			
Library_ID (PK)	Library_Name	Location	Institute_ID (FK)
L001	Central Library	Building A	I001
L002	Engineering Library	Building B	I002
L003	Management Library	Building C	I003

MEMBER			
Member_ID (PK)	Member_Name	Address	Contact_No
M001	Parth Bhayani	Pune	9876543210
M002	Ayush Bhosale	Mumbai	9123456780
M003	Siddhesh Chavan	Nashik	9988776655

AUTHOR			
Author_ID (PK)	Author_Name	Email	Contact_No
A001	C. J. Date	cjdate@gmail.com	9876500001
A002	Herbert Schildt	schildt@gmail.com	9876500002
A003	Ramesh Bangia	bangia@gmail.com	9876500003

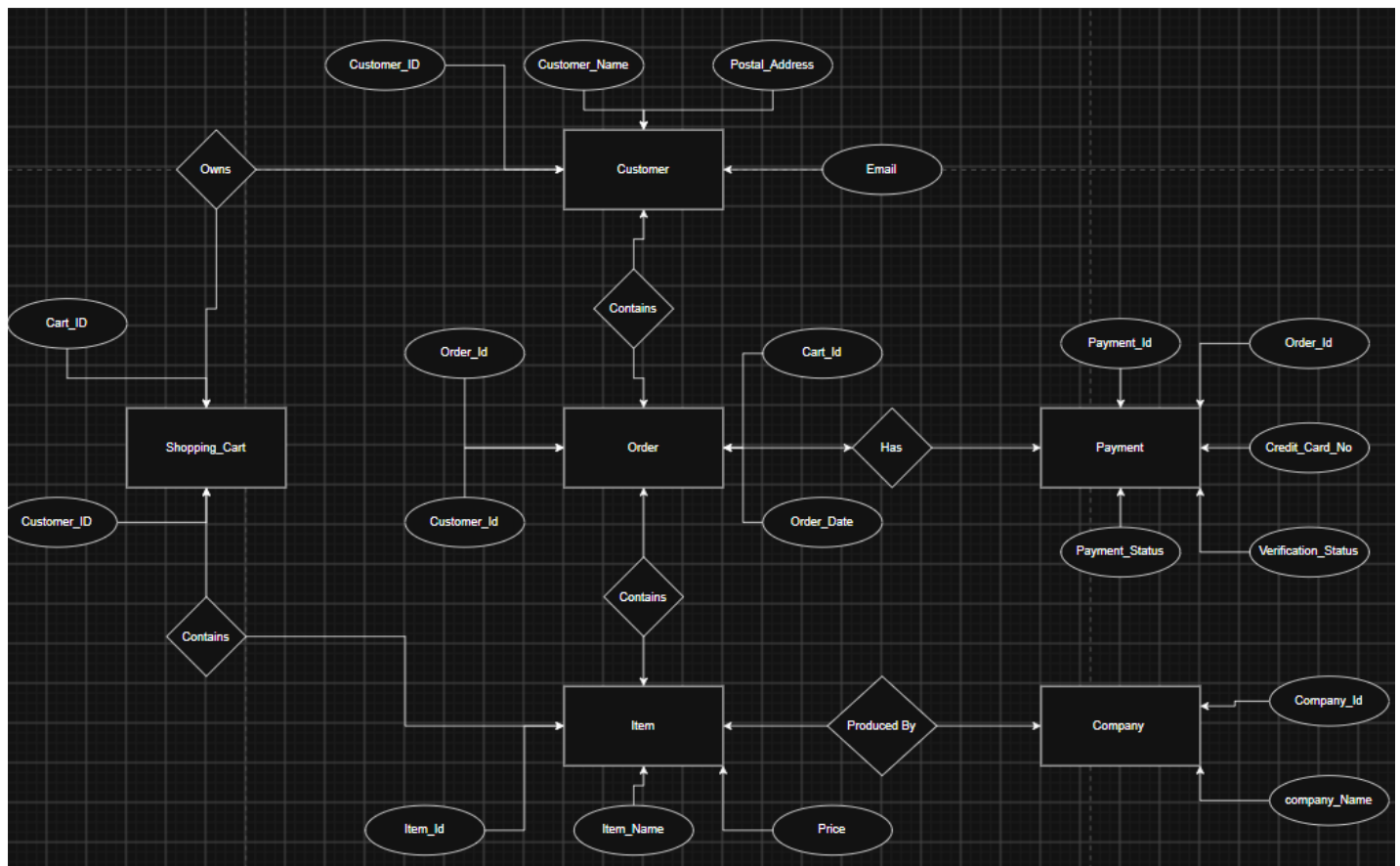
PUBLISHER		
Publisher_ID (PK)	Publisher_Name	Address
P001	McGraw Hill	New Delhi
P002	Pearson	Bangalore
P003	BPB Publications	Mumbai

SELLER			
Seller_ID (PK)	Seller_Name	Address	Publisher_ID (FK)
S001	ABC Book Store	Pune	P001
S002	Knowledge Hub	Mumbai	P002
S003	Campus Books	Pune	P003

3) Online Grocery Ordering System

- Customer must orders one or more items. Each customer is identified by his/her name. The postal address and email address of the customer are also required for the correct placement of order.
- Items have their name and price to be shown to the customers.
- Shopping carts are created by the orders of the customers.
- Shopping carts contain at least one item.
- An order must have at least one requested item.
- Not all items have customers.
- Distinct companies produce distinct items which are differentiated by the company's name.
- Payment must be made via credit cards. The payment process has to be done after the credit card's verification.

4)



Relational Schema :-

```

CUSTOMER(
    Customer_ID PK,
    Customer_Name,
    Postal_Address,
    Email
)
    
```

```
SHOPPING_CART(  
    Cart_ID PK,  
    Customer_ID FK  
)
```

```
COMPANY(  
    Company_ID PK,  
    Company_Name  
)
```

```
ITEM(  
    Item_ID PK,  
    Item_Name,  
    Price,  
    Company_ID FK  
)
```

```
ORDER(  
    Order_ID PK,  
    Customer_ID FK,  
    Cart_ID FK,  
    Order_Date  
)
```

```
PAYMENT(  
    Payment_ID PK,  
    Order_ID FK,  
    Credit_Card_No,  
    Payment_Status,  
    Verification_Status  
)
```

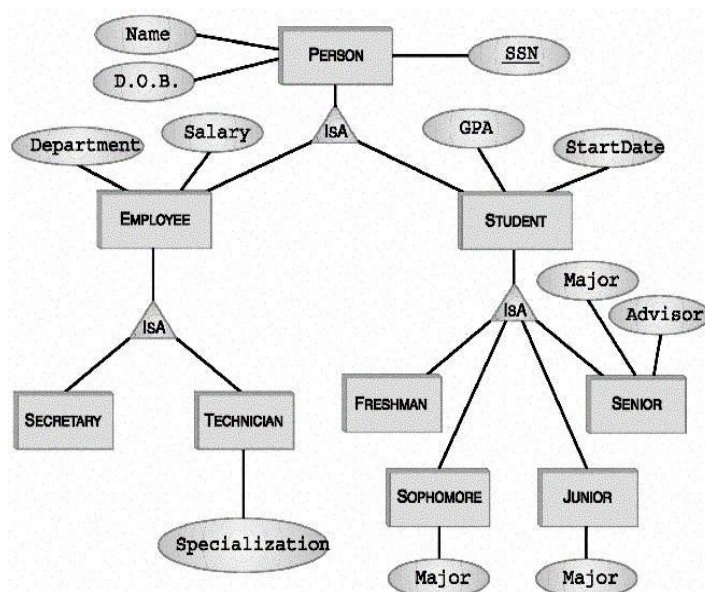
Relational Model :-

CUSTOMER				
Customer_ID (PK)	Customer_Name	Postal_Address	Email	
C001	Rahul Sharma	Pune	rahul@gmail.com	
C002	Priya Patel	Mumbai	priya@gmail.com	
C003	Amit Verma	Nashik	amit@gmail.com	
SHOPPING_CART				
Cart_ID (PK)	Customer_ID (FK)			
CT001	C001			
CT002	C002			
CT003	C003			
COMPANY				
Company_ID (PK)	Company_Name			
CO001	Fortune Foods			
CO002	Tata Consumer			
CO003	ITC Limited			
ITEM				
Item_ID (PK)	Item_Name	Price	Company_ID (FK)	
I001	Rice 5kg	350	CO001	
I002	Cooking Oil	180	CO002	
I003	Sugar 1kg	55	CO003	
ORDER				
Order_ID (PK)	Customer_ID (FK)	Cart_ID (FK)	Order_Date	
O001	C001	CT001	20-07-2026	
O002	C002	CT002	21-07-2026	
O003	C003	CT003	22-07-2026	
PAYMENT				
Payment_ID (PK)	Order_ID (FK)	Credit_Card_No	Payment_Status	Verification_Status
P001	O001	4.11111E+15	Paid	Verified
P002	O002	5.55544E+15	Pending	Pending
P003	O003	4.44433E+15	Paid	Verified

Implementation Question (EER):-

Q.1 Draw the EER for SIU library management and convert into relational model.

Q.2 Convert following EER to relational model



Input and Output:

		PERSON	
SSN (PK)		Name	DOB
	1001	Rahul Sharma	12-05-2000
	1002	Priya Patel	18-08-2001
	1003	Amit Verma	25-01-1999
EMPLOYEE			
SSN (PK/FK)		Department	Salary
	1001	Computer	55000
	1003	Mechanical	60000
STUDENT			
SSN (PK/FK)		GPA	StartDate
	1002	8.9	15-08-2024
SECRETARY			
SSN (PK/FK)			
	1001		
TECHNICIAN			
SSN (PK/FK)		Specialization	
	1003	Network Administration	
FRESHMAN			
SSN (PK/FK)			
	1004		
SOPHOMORE			
SSN (PK/FK)		Major	
	1005	Computer Engineering	
JUNIOR			
SSN (PK/FK)		Major	
	1006	Information Technology	
SENIOR			
SSN (PK/FK)		Major	Advisor
	1002	Computer Engineering	Dr. Ashwini More

Conclusion: Thus we have implemented ER and EER diagrams.